Regularity lemma for Gaussian polynomials

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Abstract

We prove a regularity lemma for polynomials with fixed degree d and any number of standard Gaussian input variables. Following modern approaches to the Szemerédi regularity lemma, we deduce our results from a general formulation of the regularity lemma using ultraproducts. Along the way, we give a direct combinatorial proof of the celebrated "fourth moment theorem" of Nualart and Peccati, which characterizes the asymptotically Gaussian polynomials inside degree-d Gaussian chaos as those such that $\mathbb{E}\, p^2 \to 1$ and $\mathbb{E}\, p^4 \to 3$.

1 Introduction

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$. How many "different" degree-d polynomials $p(X_1, \dots, X_n)$ are there, in the sense of, say, the law of p as a random variable? While p seems like it could become arbitrarily complicated as $n \to \infty$, this is not the case. Some examples are:

1. $p(X_1) = \mathrm{He}_d(X_1)$ where He_d is the degree-d probabilists' Hermite polynomial.
2. $p(X_1, \dots, X_n) = \frac{1}{\sqrt{n^d!}} \sum_{i=1}^n \mathrm{He}_d(X_i)$. By the central limit theorem, the law of this random variable is asymptotically $\mathcal{N}(0,1)$ as $n \to \infty$.
3. $p(X_1, \dots, X_n) = \frac{1}{n} \sum_{1 \le i < j \le n} X_i X_j$. This is asymptotic to $\frac{1}{2}\mathrm{He}_2(Z)$ for $Z \sim \mathcal{N}(0,1)$.

Despite the limit $n \to \infty$ in examples 2 and 3, the law of p approaches the law of a polynomial in just one standard Gaussian variable. Herry et al. [HMP24] proved that this is a general phenomenon: any sequence of degree-d Gaussian polynomials which converges in law, converges in law towards another Gaussian polynomial (possibly of lower degree).

Here we characterize the asymptotic limits of Gaussian polynomials in much stronger ways, broadly termed "regularity lemmas for Gaussian polynomials". First, we prove a regularity lemma in the traditional sense of approximation by a structured constant-size object. For a set of variables X, let $\mathcal{H}_d(X)$ be the span of the degree-d Hermite polynomials in X, and let $\mathcal{H}_{\le d}(X)$ be the span of all monomials in X up to degree d. $\mathcal{H}_d(X)$ is also called the "degree-d Gaussian chaos" for X.

Theorem 1.1 (Regularity lemma, approximation version)**.** *For every $d \in \mathbb{N}$, $\varepsilon > 0$ there exists $M = M(d, \varepsilon)$ such that: for all $p \in \mathcal{H}_{\le d}(X)$ such that $\mathbb{E}\, p^2 \le 1$, there is $m \le M$, a polynomial $q(Z_1, \dots, Z_m), \deg(q) \le d$ and $Z_a \in \mathcal{H}_{\le d}(X)$ such that:*

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(i) $\mathbb{E}(p-q)^2 \le \varepsilon$

(ii) $W_2((Z_1,\dots,Z_m),\mathcal{N}(0,1)^{\otimes m}) \le \varepsilon$ *where* W_2 *is 2-Wasserstein distance.*

To prove Theorem 1.1, we will deduce it from a semantically stronger theorem characterizing the *ultraproduct* of Gaussian polynomials [ES12, Tao13]. Roughly speaking, our theorem states that the possible limits of Gaussian polynomials are "graded Gaussian polynomials".

At an intermediate level between the completely finitary Theorem 1.1 and the ultraproduct viewpoint, one can also view a regularity lemma as a "compactification". For example, the space of graphons compactifies the space of graphs under the cut metric [LS06, ES12, Lov12]. At this level, we prove that graded-degree-d Gaussian polynomials are the closure of degree-d Gaussian polynomials in distribution (and the compactification when the space is bounded, $\mathbb{E}\,p^2 \le 1$).

Theorem 1.2 (Regularity lemma, compactification version)**.** *Let X be a set of countably many variables, and let Z be a set of graded variables with countably many variables of each grade. Let $\mathcal{H}_d^{\mathrm{gr}}(Z)$ be the span of graded-degree-d Hermite polynomials in Z. Let $d \in \mathbb{N}$.*

(i) For all $p \in \mathcal{H}_d^{\mathrm{gr}}(Z)$, there exists a sequence $p_n \in \mathcal{H}_d(X_1,\dots,X_n)$ such that $p_n \xrightarrow{d} p$.

(ii) If $p_n \in \mathcal{H}_d^{\mathrm{gr}}(Z)$ satisfies $p_n \xrightarrow{d} p$ for some p, then $p_n \xrightarrow{d} p'$ for some $p' \in \mathcal{H}_d^{\mathrm{gr}}(Z)$.

Equivalently, $\{\mathrm{Law}(p) : p \in \mathcal{H}_d^{\mathrm{gr}}(Z)\}$ is the closure of $\{\mathrm{Law}(p) : p \in \mathcal{H}_d(X)\}$ in the space of probability measures on $\mathbb{R}$ with the weak topology. A consequence of (ii) is that $\{\mathrm{Law}(p) : p \in \mathcal{H}_d^{\mathrm{gr}}(Z), \mathbb{E}\,p^2 \le 1\}$ is compact in the weak topology.

In the main text, we will prove multivariate versions of these theorems. Using a hypercontractivity argument, we can also generically upgrade Theorem 1.2 from the weak topology to Wasserstein metrics, see Section A. Considering it to be a problem of independent interest, Herry et al. [HMP24] proved a variant of this theorem, which is that $\mathcal{H}_{\le d}(X)$ is closed in distribution when X is a set of countably many standard Gaussian variables.

There is a large body of work on characterizing the Gaussian limits inside $\mathcal{H}_d(X)$. The famous "fourth moment theorem" of Nualart and Peccati [NP05, NOL08, NP15] states that, if $\mathbb{E}\,p_n^2 \to 1$, then $p_n \xrightarrow{d} \mathcal{N}(0,1)$ if and only if $\mathbb{E}\,p_n^4 \to 3$ if and only if all "contraction norms" of p_n vanish. We identify a small extension: if and only if all proper flattenings of the coefficient tensor of p_n have vanishing spectral norm. We use this to give a direct combinatorial proof of the fourth moment theorem and then prove the regularity lemma.

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2 Graded Gaussian polynomials

A graded variable is a variable with an integer label $r > 0$ called its "grade". We write $X_1, X_2, X_3, \dots$ for "original" grade-one variables and $Z_1^{(r)}, Z_2^{(r)}, Z_3^{(r)}, \dots$ for grade-r variables, with the identifica-

tion $X_i = Z_i^{(1)}$ at grade one. Sample $Z_i^{(r)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ independently across all variables regardless of grade.

The idea of the asymptotic limit is that an asymptotically Gaussian random variable in $\mathcal{H}_d(X)$ can be replaced in the limit by a single new grade-d variable. In this way, we will represent the limit of a sequence of Gaussian polynomials by a single graded polynomial. When we do an operation on this limit object which depends on the degree (e.g. calculate the degree-d weight) then we use the grade information to recover the "original" degree.

We use the following standard notions from Gaussian analysis. Let He_k be the unnormalized probabilists' Hermite polynomials. Let $\psi_k = \mathrm{He}_k / \sqrt{k!}$ be their normalizations. For a finitely-supported multi-index $\alpha \in \mathbb{N}^X$ let $X^\alpha = \prod_i X_i^{\alpha_i}$ and similarly for He_α and ψ_α. Let $|\alpha| = \sum_i \alpha_i$ and $\alpha! = \prod_i \alpha_i!$. Then $\{\psi_\alpha(X) : \alpha \in \mathbb{N}^X, |\alpha| < \infty\}$ is an orthonormal basis of $L_2(X)$. Let $\mathcal{H}_d(X)$ be the closed span of the degree-d Hermite polynomials in X. Let $\mathcal{H}_{\le d}(X) = \oplus_{i=0}^d \mathcal{H}_i(X)$. Let $\langle f, g\rangle = \mathbb{E}\, fg$ and $\|f\|_2 = (\mathbb{E}\, f^2)^{1/2}$.

Polynomials in Z have an additional notion of total grade, or graded degree. Let $\mathrm{gr}(\alpha) = \sum_{i,r} r\alpha_{ir}$ be the total grade of a multi-index $\alpha \in \mathbb{N}^Z$. The Hermite polynomial $\mathrm{He}_\alpha(Z)$ is defined as before, and $\psi_\alpha(Z)$ are still an orthonormal basis for $L_2(Z)$. Define graded-degree-d chaos by

$$\mathcal{H}_d^{\mathrm{gr}}(Z) = \overline{\mathrm{span}}\,(\mathrm{He}_\alpha(Z) : \mathrm{gr}(\alpha) = d)\ . \tag{2.1}$$

The notation $\overline{\mathrm{span}}$ denotes the closure of the linear span in $L_2(Z)$.

The following notions on Gaussian space interact with degree, and so behave differently on graded Gaussian space. Let Π_d be the operator on $L_2(Z)$ which projects to $\mathcal{H}_d^{\mathrm{gr}}(Z)$. Let $W_d(f) = \|\Pi_d f\|_2^2$. For variables with grade r, the Ornstein–Uhlenbeck noise operator at parameter $t \in [0,1]$ acts by sampling standard Gaussians W with covariance $\mathbb{E}[W_i^{(r)} Z_i^{(r)}] = t^r$. Under these definitions, we still have the noise correlation and Parseval identities,

$$\mathop{\mathbb{E}}_{(W,Z)\sim t\text{-correlated}}[f(W)g(Z)] = \sum_{d\ge 0} t^d \langle \Pi_d f, \Pi_d g\rangle\,, \qquad \mathbb{E}[f(Z)^2] = \sum_{d\ge 0} W_d(f)\,. \tag{2.2}$$

We verify that all $p \in \mathcal{H}_d^{\mathrm{gr}}(Z)$ can be approximated by finite Gaussian polynomials, which is the first (easier) part of Theorem 1.2.

Lemma 2.1. *Let $d_k \in \mathbb{N}$ for $k \in \mathbb{N}$. For all $p_k \in \mathcal{H}_{d_k}^{\mathrm{gr}}(Z)$ there exist sequences $p_{n,k} \in \mathcal{H}_{d_k}(X_1, \dots, X_n)$ such that $(p_{n,k})_{k\in\mathbb{N}} \xrightarrow{d} (p_k)_{k\in\mathbb{N}}$.*

Proof. For each graded variable $Z_i^{(r)}$ introduce new standard Gaussians $X_{ir,j}$ for $j \in [N]$. Define approximants for $Z_i^{(r)}$ and $\mathrm{He}_m(Z_i^{(r)})$ by,

$$S_{ir} = \frac{1}{\sqrt{N}} \sum_{j=1}^N \psi_r(X_{ir,j}), \qquad U_{m,ir} = \frac{1}{N^{m/2}} \sum_{\substack{j_1,\dots,j_m\in[N]\\ j_1,\dots,j_m \text{ distinct}}} \psi_r(X_{ir,j_1})\psi_r(X_{ir,j_2})\cdots\psi_r(X_{ir,j_m})\,.$$

We approximate $p_k \in \mathcal{H}_{d_k}^{\mathrm{gr}}(Z)$ by:

$$p_{N,k} = \sum_\alpha c_\alpha \prod_{i,r} U_{\alpha_{ir},ir} \qquad \text{where} \qquad p_k = \sum_\alpha c_\alpha \,\mathrm{He}_\alpha(Z) = \sum_\alpha c_\alpha \prod_{i,r} \mathrm{He}_{\alpha_{ir}}(Z_i^{(r)})\,.$$

All terms in $p_{N,k}$ have Hermite degree d_k so $p_{N,k} \in \mathcal{H}_{d_k}(X)$. It remains to show $(p_{N,k})_{k\in\mathbb{N}} \xrightarrow{d} (p_k)_{k\in\mathbb{N}}$ as $N \to \infty$. This is equivalent to $(p_{N,1}, \dots, p_{N,k}) \xrightarrow{d} (p_1, \dots, p_k)$ for every finite k.

First assume each $p_k = \mathrm{He}_{\alpha^k}(Z)$ is a single Hermite polynomial. Consider $\tilde{p}_{N,k} = \mathrm{He}_{\alpha^k}(S)$. By the multivariate central limit theorem, $(\tilde{p}_{N,1}, \dots, \tilde{p}_{N,k}) \xrightarrow{d} (p_1, \dots, p_k)$. On the other hand,

$$\mathbb{E}(\tilde{p}_{N,k} - p_{N,k})^2 = O(N^{-1})\,. \tag{2.3}$$

This is a combinatorial moment calculation which we include in Lemma B.5.

Combine Eq. (2.3) and the convergence $(\tilde{p}_{N,1}, \dots, \tilde{p}_{N,k}) \xrightarrow{d} (p_1, \dots, p_k)$ with Slutsky's lemma to deduce $(p_{N,1}, \dots, p_{N,k}) \xrightarrow{d} (p_1, \dots, p_k)$. This completes the case of single Hermite polynomials.

The continuous mapping theorem gives $(p_{N,1}, \dots, p_{N,k}) \xrightarrow{d} (p_1, \dots, p_k)$ if every p_k is a finite sum. For $p_k \in \mathcal{H}^{\mathrm{gr}}_{d_k}(Z)$ with an infinite number of terms, take a sequence of finite L_2 approximants, approximate each one, then take a diagonal sequence to obtain $(p_{N,1}, \dots, p_{N,k}) \xrightarrow{d} (p_1, \dots, p_k)$. □

3 Proof of the regularity lemma

3.1 Regularity lemma for degree 1 and 2

A degree-1 polynomial is of the form $p(X) = \sum_i c_i X_i$. The law of this polynomial is always Gaussian, so the characterization of degree-1 Gaussian polynomials is immediate.

As a warm-up, we prove a version of Theorem 1.1 for degree-2 Gaussian polynomials.

Lemma 3.1. *Let $\varepsilon > 0$. For all $p \in \mathcal{H}_2(X_1, \dots, X_n)$ such that $\mathbb{E}\, p^2 \le 1$, there exist $Z_0 \in \mathcal{H}_2(X_1, \dots, X_n)$ and $Z_1, \dots, Z_m \in \mathcal{H}_1(X_1, \dots, X_n)$ with $m \le 2\varepsilon^{-2}$ and coefficients λ_a such that:*

(i) $p = Z_0 + \sum_{a=1}^m \lambda_a \mathrm{He}_2(Z_a)$.

(ii) $Z_0, Z_1, \dots, Z_m$ *are independent and* $W_2(Z_0, \mathcal{N}(0,s)) \le C\varepsilon$ *for an absolute constant C where* $s = \mathbb{E}\, Z_0^2$.

Proof. Write p in the Hermite basis as $p(X) = \sum_{i\ne j} P_{ij} X_i X_j + \sum_{i=1}^n P_{ii} \mathrm{He}_2(X_i)$ for a symmetric coefficient matrix $P \in \mathbb{R}^{n\times n}_{\mathrm{sym}}$. Let $\sum_a \lambda_a v_a v_a^\intercal$ be the spectral decomposition of P. Then

$$p(X) = X^\intercal P X - \mathrm{tr}(P) = \sum_a \lambda_a \langle v_a, X\rangle^2 - \mathrm{tr}(P) = \sum_a \lambda_a(\langle v_a, X\rangle^2 - 1) = \sum_a \lambda_a \mathrm{He}_2(\langle v_a, X\rangle)\,. \tag{3.1}$$

By the orthogonality of the eigendecomposition, $\langle v_a, X\rangle$ are independent standard Gaussians.

We have $\frac{1}{2}\mathbb{E}\, p^2 = \sum_{i,j=1}^n P_{ij}^2 =: \|P\|_F^2$. On the other hand, $\|P\|_F^2 = \sum_a \lambda_a^2$. Since $\mathbb{E}\, p^2 \le 1$, we deduce that $\sum_a \lambda_a^2 \le \frac{1}{2}$.

Let $A = \{a : |\lambda_a| \ge \varepsilon/2\}$. By the above, $|A| \le 2\varepsilon^{-2}$. Write

$$p = \sum_{a\in A} \lambda_a \mathrm{He}_2(\underbrace{\langle v_a, X\rangle}_{=:Z_a}) + \underbrace{\sum_{a\notin A} \lambda_a \mathrm{He}_2(\langle v_a, X\rangle)}_{=:Z_0}\,.$$

Since the $\mathrm{He}_2(\langle v_a, X\rangle)$ are independent and mean-zero, a central limit theorem implies Z_0 is approximately Gaussian if there are many summands. If there are few summands, Z_0 is small since $|\lambda_a| \le \varepsilon/2$. Both cases are handled simultaneously by the CLT from [Rio09, Theorem 4.1] which yields for an absolute constant C, with $s = \mathbb{E}\, Z_0^2 = 2\sum_{a\notin A} \lambda_a^2$,

$$W_2(Z_0, \mathcal{N}(0,s)) \le \frac{C}{\sqrt{s}} \left(\sum_{a\notin A} \lambda_a^4\right)^{1/2} \le C \max_{a\notin A} |\lambda_a| \le C\varepsilon\,. \qquad \square$$

3.2 Preliminaries on ultraproducts

A collection $\mathcal{U}$ of subsets of $\mathbb{N}$ is a *non-principal ultrafilter* if:

(i) $\mathcal{U}$ contains $\mathbb{N}$ but not $\emptyset$.

(ii) If $A \in \mathcal{U}$ and $A \subseteq B$, then $B \in \mathcal{U}$.

(iii) If $A, B \in \mathcal{U}$ then $A \cap B \in \mathcal{U}$

(iv) If $A \subseteq \mathbb{N}$ then exactly one of A and $\mathbb{N} \setminus A$ is in $\mathcal{U}$.

(v) There are no finite sets in $\mathcal{U}$.

We say that $A \in \mathcal{U}$ is a $\mathcal{U}$-*large* set of natural numbers.

Fix any[1] non-principal ultrafilter $\mathcal{U}$ on $\mathbb{N}$. We can use $\mathcal{U}$ to "pick" the limit of a sequence of objects inside a compact Hausdorff space, as summarized in the next lemma.

Lemma 3.2 ([Wil70, Theorems 13.7 and 17.4])**.**

(i) (Existence and uniqueness of limits) Let S be a compact Hausdorff space. For all sequences $x_n \in S$ there exists a unique $x \in S$ such that, for all open neighborhoods V of x, we have $\{n : x_n \in V\} \in \mathcal{U}$. Write $x = \lim_{n\to\mathcal{U}} x_n$ or $x_n \underset{n\to\mathcal{U}}{\longrightarrow} x$.

(ii) (Continuity of limits) Let S, T be Hausdorff spaces and $f : S \to T$ continuous. If $x_n \underset{n\to\mathcal{U}}{\longrightarrow} x$ in S then $f(x_n) \underset{n\to\mathcal{U}}{\longrightarrow} f(x)$ in T.

(iii) (Extraction of a subsequence) Let $x_n \underset{n\to\mathcal{U}}{\longrightarrow} x$ in a first-countable Hausdorff space S (in particular, a metrizable space). Then there exists an infinite subsequence such that $x_n \to x$ in S along the subsequence. Moreover, the subsequence may be chosen inside any $A \in \mathcal{U}$.

While Lemma 3.2 applies to compact spaces, sequences inside a non-compact space can still be studied using *ultraproducts*. We are specifically interested in sequences and limits of Gaussian polynomials. Define

$$B_d = \{(p_n)_{n\in\mathbb{N}} : p_n \in \mathcal{H}_{\le d}(X_1, \dots, X_n),\ \sup_n \mathbb{E}\, p_n^2 < \infty\}\,, \qquad N_d = \{(p_n)_{n\in\mathbb{N}} \in B_d : \lim_{n\to\mathcal{U}} \mathbb{E}\, p_n^2 = 0\}\,.$$

[1] A non-principal ultrafilter $\mathcal{U}$ can be easily constructed using Zorn's lemma, which uses the axiom of choice (ZFC). We consider it all but certain that Theorems 3.8 and 3.9 can also be proven without choice.

The scalar ultralimit in the definition of N_d exists because $\mathbb{E}\, p_n^2$ is bounded and so $(\mathbb{E}\, p_n^2)_{n\in\mathbb{N}}$ lies in a compact subset of $\mathbb{R}$. By the triangle inequality, N_d is a linear subspace of B_d. The *Hilbert ultraproduct* is the vector space

$$\mathscr{H}_{\leq d} = B_d/N_d\,.$$

We write $[p_n]$ for elements of $\mathscr{H}_{\leq d}$ which formally are equivalence classes of the form $[p_n] = \{(p_n + r_n)_{n\in\mathbb{N}} : (r_n)_{n\in\mathbb{N}} \in N_d\}$. A crucial property of $\mathscr{H}_{\leq d}$ is that it inherits the Hilbert space structure from $\mathcal{H}_{\leq d}(X)$.

Lemma 3.3. *$\mathscr{H}_{\leq d}$ is a Hilbert space with inner product $\langle [p_n], [q_n]\rangle = \lim_{n\to\mathcal{U}} \mathbb{E}\, p_n q_n$.*

Proof. It is easy to check that $\langle [p_n], [q_n]\rangle$ defines a valid inner product on $\mathscr{H}_{\leq d} = B_d/N_d$. What remains is to check that the space is complete.

Consider the space B_d with the supremum norm $\|(p_n)_{n\in\mathbb{N}}\|_\infty = \sup_n \|p_n\|_2$. Since each coordinate space $\mathcal{H}_{\leq d}(X_1, \ldots, X_n)$ is complete, B_d is complete in this norm (B_d is a Banach space). Moreover, $(p_n)_{n\in\mathbb{N}} \mapsto \lim_{n\to\mathcal{U}} \|p_n\|_2$ is 1-Lipschitz in this norm, so the kernel N_d is closed in this norm. Thus B_d/N_d is complete in the quotient norm $\|[p_n]\|_{B_d/N_d} := \inf_{(r_n)_{n\in\mathbb{N}}\in N_d} \|(p_n + r_n)_{n\in\mathbb{N}}\|_\infty$.

It remains to show that the quotient norm agrees with the inner-product norm $\sqrt{\langle [p_n], [p_n]\rangle} = \lim_{n\to\mathcal{U}} \|p_n\|_2$ which will prove that $\mathscr{H}_{\leq d}$ is complete in the inner-product norm. Set $L = \lim_{n\to\mathcal{U}} \|p_n\|_2$. Every representative $(q_n)_{n\in\mathbb{N}} \in B_d$ of $[p_n]$ satisfies $\lim_{n\to\mathcal{U}} \|q_n\|_2 = L$. Hence $\sup_n \|q_n\|_2 \geq L$ by Lemma 3.2(iii) and therefore the quotient norm is at least L.

Conversely, for every $\varepsilon > 0$, the set $A = \{n : \|p_n\|_2 < L + \varepsilon\}$ belongs to $\mathcal{U}$ by Lemma 3.2(i). Define $q_n = p_n$ for $n \in A$ and $q_n = 0$ otherwise. Then $(p_n - q_n) \in N_d$ and $\sup_n \|q_n\|_2 \leq L + \varepsilon$. Therefore the quotient norm is at most $L + \varepsilon$. Letting $\varepsilon \to 0$ shows that it equals L, completing the proof. □

Let $\mathscr{H}_d = \{[p_n] : \forall n \in \mathbb{N}.\ p_n \in \mathcal{H}_d(X)\}$ be the degree-d component. Since $\|\Pi_k p\|_2 \leq \|p\|_2$ then N_d vanishes under Π_k and Π_k is well-defined on $\mathscr{H}_{\leq d}$. Hence, we maintain the decomposition $\mathscr{H}_{\leq d} = \oplus_{j=0}^d \mathscr{H}_j$.

Two remarks on the purpose of introducing the ultraproduct:

1. A technical advantage of the ultralimit is that it consistently chooses a limit inside all compact spaces simultaneously. If x_n are in a compact Hausdorff space (e.g. bounded real numbers) then we can treat $\lim_{n\to\mathcal{U}} x_n$ like a normal limit and delay applying compactness, passing to subsequences, etc. until we are ready to apply Lemma 3.2(iii) to "exit" the ultraproduct.

2. For sequences in a non-compact space, ontologically, we can think of $[p_n] \in \mathscr{H}_{\leq d}$ as a new type of object representing the limit of the sequence p_n. These objects inherit all of the usual operations of the original space, for example, given $[p_n], [q_n] \in \mathscr{H}_{\leq d}$ we can select representatives p_n, q_n and form $[p_n + q_n] \in \mathscr{H}_{\leq d}$. Classifying the limits of Gaussian polynomials reduces to classifying the elements of $\mathscr{H}_{\leq d}$ (starting with Lemma 3.3 which shows they form a Hilbert space).

   In the asymptotic Hilbert space, asymptotic approximations become exact relations. By Lemma 3.3,

$$\|x_n - y_n\|_2 \underset{n\to\mathcal{U}}{\longrightarrow} 0 \text{ if and only if } [x_n] = [y_n] \text{ in } \mathscr{H}_{\leq d}$$
$$\langle x_n, y_n\rangle \underset{n\to\mathcal{U}}{\longrightarrow} 0 \text{ if and only if } [x_n] \perp [y_n] \text{ in } \mathscr{H}_{\leq d}\,.$$

3.3 Orthogonal decomposition of the asymptotic Hilbert space

Since $\mathscr{H}_d$ is a Hilbert space, by Zorn's lemma it has an orthonormal basis, which is uncountable. The goal of this section is to construct an orthonormal basis of "graded Hermite polynomials" for $\mathscr{H}_d$ which will solve, in a way, the problem of describing $\mathscr{H}_d$.

For $p \in \mathcal{H}_r(X)$ and $q \in \mathcal{H}_s(X)$, define the *top product* (or *Wick product*)

$$p \diamond q = \Pi_{r+s}(pq).$$

It can be seen that $\diamond$ is commutative and associative. We can also lift the top product to $\mathscr{H}_d$.

Lemma 3.4. *$\diamond$ is a well-defined and continuous bilinear map $\mathscr{H}_r \times \mathscr{H}_s \to \mathscr{H}_{r+s}$.*

Proof. By contractivity of projection, Cauchy–Schwarz, and hypercontractivity (Lemma A.3),

$$\|p \diamond q\|_2 \le \|pq\|_2 \le \|p\|_4 \|q\|_4 \le 3^{(r+s)/2} \|p\|_2 \|q\|_2.$$

Consequently, if $p_n \in \mathcal{H}_r(X)$ and $q_n \in \mathcal{H}_s(X)$ satisfy $\sup_n \mathbb{E}\, p_n^2 < \infty$ and $\sup_n \mathbb{E}\, q_n^2 < \infty$, then $\sup_n \mathbb{E}(p_n \diamond q_n)^2 < \infty$ and hence $(p_n \diamond q_n)_{n\in\mathbb{N}} \in B_{r+s}$. Lift $\diamond$ to $\mathscr{H}_r \times \mathscr{H}_s$ by $[p_n] \diamond [q_n] := [p_n \diamond q_n]$. To check that $\diamond$ is independent of the representatives, suppose $[p_n] = [p_n']$ and $[q_n] = [q_n']$. By bilinearity and the preceding bound,

$$\|p_n \diamond q_n - p_n' \diamond q_n'\|_2 \le 3^{(r+s)/2} \big(\|p_n - p_n'\|_2 \|q_n\|_2 + \|p_n'\|_2 \|q_n - q_n'\|_2 \big).$$

The right-hand side tends to zero along $\mathcal{U}$, since $\|q_n\|_2, \|p_n'\|_2$ are bounded and the difference terms tend to zero along $\mathcal{U}$. Thus the product is well-defined.

Finally, taking ultralimits in the first inequality gives

$$\|u \diamond v\|_{\mathscr{H}_{r+s}} \le 3^{(r+s)/2} \|u\|_{\mathscr{H}_r} \|v\|_{\mathscr{H}_s}.$$

Hence $\diamond$ is a bounded bilinear map and therefore jointly continuous. □

Let $\mathscr{D}_d$ be the subspace of $\mathscr{H}_d$ generated by lower-degree polynomials,

$$\mathscr{D}_d = \overline{\operatorname{span}}\{u \diamond v : u \in \mathscr{H}_r,\ v \in \mathscr{H}_{d-r},\ 0 < r < d\}. \tag{3.2}$$

Since $\mathscr{H}_d$ is a Hilbert space, we can form the orthogonal decomposition $\mathscr{H}_d = \mathscr{D}_d \oplus \mathscr{P}_d$ where $\mathscr{P}_d$ is the orthogonal complement of $\mathscr{D}_d$. We call the elements of $\mathscr{P}_d$ *primitive*.

Select an arbitrary orthonormal basis $\{[e_{n,i}] : i \in I_d\}$ for $\mathscr{P}_d$ and let $I = \sqcup_{d\in\mathbb{N}} I_d$. We also normalize $\mathbb{E}\, e_{n,i}^2 = 1$ exactly. For finitely-supported $\alpha \in \mathbb{N}^I$ define $\operatorname{gr}(\alpha) = \sum_{d\in\mathbb{N}} \sum_{i\in I_d} d\alpha_i$. By the definition of $\mathscr{D}_d$ and $\mathscr{P}_d$ there is a dense spanning set for $\mathscr{H}_d$ given by

$$\left\{ \frac{1}{\sqrt{\alpha!}} \bigdiamond_{i\in I} [e_{n,i}^{\diamond \alpha_i}] : \alpha \in \mathbb{N}^I, \operatorname{gr}(\alpha) = d \right\}.$$

We will prove that all primitive elements are asymptotically Gaussian, and these top products are asymptotically orthonormal Hermite polynomials. This will characterize the basis for $\mathscr{H}_d$ as the collection of graded Hermite polynomials in the primitive elements.

Theorem 3.5 (Regularity lemma, ultrapower version)**.** *$\{[\psi_\alpha(e_n)] : \alpha \in \mathbb{N}^I, \mathrm{gr}(\alpha) = d\}$ is an orthonormal basis for $\mathscr{H}_d$. Furthermore, let $(Z_i)_{i\in I}$ be independent standard Gaussian variables such that I_d is assigned grade d. Let $d_k \in \mathbb{N}$ for $k \in \mathbb{N}$. For any $[p_{n,k}] \in \mathscr{H}_{d_k}$ write the basis representation of $[p_{n,k}]$ and define $p_k \in \mathcal{H}^{\mathrm{gr}}_{d_k}(Z)$ by*

$$[p_{n,k}] = \sum_{\alpha\in\mathbb{N}^I:\mathrm{gr}(\alpha)=d_k} c_{k,\alpha}[\psi_\alpha(e_n)]\,, \qquad p_k = \sum_{\alpha\in\mathbb{N}^I:\mathrm{gr}(\alpha)=d_k} c_{k,\alpha}\psi_\alpha(Z)\,.$$

Then $(p_{n,k})_{k\in\mathbb{N}} \xrightarrow[n\to\mathcal{U}]{d} (p_k)_{k\in\mathbb{N}}$.

Note that the target $(p_k)_{k\in\mathbb{N}}$ of the convergence in distribution is independent of the choice of representatives $p_{n,k}$ for $[p_{n,k}]$ by Slutsky's lemma applied to the difference element from N_d.

3.3.1 Proof of Theorem 3.5

For an ordered tuple $\mathbf{i} = (i_1, \ldots, i_d)$, let $\alpha(\mathbf{i})_j = |\{s : i_s = j\}|$ and set $\mathrm{He}_{i_1\cdots i_d}(X) := \mathrm{He}_{\alpha(\mathbf{i})}(X)$. For $p \in \mathcal{H}_d(X_1, \ldots, X_n)$, the *Hermite coefficient tensor* $P \in (\mathbb{R}^n)^{\otimes d}_{\mathrm{sym}}$ is the symmetric order-d tensor with dimension n such that,

$$p(X) = \frac{1}{\sqrt{d!}} \sum_{i_1,\ldots,i_d=1}^{n} P_{i_1\cdots i_d}\,\mathrm{He}_{i_1\cdots i_d}(X)\,. \tag{3.3}$$

In Nualart and Peccati's notation, p is realized as a d-dimensional multiple Wiener–Itô integral, $p = \frac{I_d(P)}{\sqrt{d!}}$ [NP05]. The entry normalization creates an important isometry between $\mathcal{H}_d(X_1, \ldots, X_n)$ and $(\mathbb{R}^n)^{\otimes d}_{\mathrm{sym}}$:

$$\mathbb{E}\,pq = \langle P, Q\rangle_F =: \sum_{i_1,\ldots,i_d} P_{i_1\cdots i_d}Q_{i_1\cdots i_d}, \qquad \mathbb{E}\,p^2 = \|P\|_F^2 := \sum_{i_1,\ldots,i_d} P^2_{i_1\cdots i_d}\,. \tag{3.4}$$

We call $P \in (\mathbb{R}^n)^{\otimes d}$ a non-symmetric Hermite coefficient tensor for p if it satisfies Eq. (3.3). If at least one tensor P, Q is symmetric in (3.4), the equality continues to hold since the other tensor can be symmetrized without loss of generality.

For $u \in \mathcal{H}_r(X)$ and $v \in \mathcal{H}_s(X)$ with Hermite coefficient tensors U, V we observe that there is a non-symmetric Hermite coefficient tensor W for $u \diamond v \in \mathcal{H}_{r+s}(X)$ given by

$$W = c_{r,s}U \otimes V, \qquad c_{r,s} = \sqrt{\frac{(r+s)!}{r!s!}}\,. \tag{3.5}$$

The factor $c_{r,s}$ accounts for normalization in (3.3).

Given a tensor P and a partition of the indices $[d]$ into two ordered sets (L, R), the *flattening* $P^{(L,R)} \in \mathbb{R}^{n^{|L|}\times n^{|R|}}$ is the matrix with entries $P^{(L,R)}_{i_L i_R} = P_{i_1\cdots i_d}$ where $i_L = (i_\ell)_{\ell\in L}$ and $i_R = (i_r)_{r\in R}$. An order-d symmetric tensor has only $d+1$ distinct flattenings. We write $P^{(k)} \in \mathbb{R}^{n^k\times n^{d-k}}$ for the kth flattening. vec denotes flattening the tensor into a column vector.

Lemma 3.6. *Let $[p_n] \in \mathscr{H}_d$. Let P_n be the Hermite coefficient tensor of p_n. Then*

$$[p_n] \in \mathscr{P}_d \quad\Longleftrightarrow\quad \lim_{n\to\mathcal{U}} \|P_n^{(r)}\| = 0 \quad \textit{for every } 0 < r < d.$$

Proof. Let $[p_n] \in \mathscr{H}_d(X), [u_n] \in \mathscr{H}_r(X), [v_n] \in \mathscr{H}_{d-r}(X)$ have Hermite coefficient tensors P_n, U_n, V_n. By the tensor isometry (3.4) and (3.5),

$$\begin{aligned}\mathbb{E}\, p_n(u_n \diamond v_n) &= c_{r,d-r}\langle P_n, U_n \otimes V_n\rangle = c_{r,d-r}\mathrm{vec}(U_n)^\intercal P_n^{(r)}\mathrm{vec}(V_n)\\ |\,\mathbb{E}\, p_n(u_n \diamond v_n)| &\le c_{r,d-r}\|P_n^{(r)}\|\|U_n\|_F\|V_n\|_F = c_{r,d-r}\|P_n^{(r)}\|\|u_n\|_2\|v_n\|_2\,.\end{aligned}$$

Suppose that $\lim_{n\to\mathcal{U}}\|P_n^{(r)}\| = 0$. Taking ultralimits and using that $\|u_n\|_2, \|v_n\|_2$ are bounded, then $\langle [p_n], [u_n \diamond v_n]\rangle_{\mathscr{H}_d} = \lim_{n\to\mathcal{U}}\mathbb{E}\, p_n(u_n \diamond v_n) = 0$. Therefore, $[p_n] \perp \mathscr{D}_d$ in this case.

Conversely, suppose $[p_n] \in \mathscr{P}_d$ and $\lim_{n\to\mathcal{U}}\|P_n^{(r)}\| > 0$. Choose unit-norm singular vectors $\tilde{U}_n, \tilde{V}_n$ for each n such that

$$(\tilde{U}_n)^\intercal P_n^{(r)}\tilde{V}_n = \|P_n^{(r)}\|\,.$$

Unflatten $\tilde{U}_n, \tilde{V}_n$ into tensors U_n, V_n. Note that since P_n is a symmetric tensor, its nonzero singular vectors also correspond to symmetric tensors. Then U_n, V_n represent unit-norm $u_n \in \mathcal{H}_r(X)$, $v_n \in \mathcal{H}_{d-r}(X)$ with,

$$\mathbb{E}\, p_n(u_n \diamond v_n) = c_{r,d-r}\langle P_n, U_n \otimes V_n\rangle = c_{r,d-r}(\tilde{U}_n)^\intercal P_n^{(r)}\tilde{V}_n = c_{r,d-r}\|P_n^{(r)}\|\,.$$

Taking the ultralimit shows $\langle [p_n], [u_n] \diamond [v_n]\rangle > 0$ contrary to $[p_n] \perp \mathscr{D}_d$. □

Lemma 3.7. *For all finitely-supported $\alpha \in \mathbb{N}^I$,*

$$\frac{1}{\sqrt{\alpha!}} \bigdiamond_{i\in I} [e_{n,i}^{\diamond\alpha_i}] = [\psi_\alpha(e_n)]\,.$$

Proof. Let $h_{n,\alpha} = \frac{1}{\sqrt{\alpha!}} \bigdiamond_{i\in I} e_{n,i}^{\diamond\alpha_i}$. We have,

$$\mathbb{E}(\psi_\alpha(e_n) - h_{n,\alpha})^2 = \mathbb{E}\,\psi_\alpha(e_n)^2 - 2\,\mathbb{E}\,\psi_\alpha(e_n)h_{n,\alpha} + \mathbb{E}\,h_{n,\alpha}^2 = \mathbb{E}\,\psi_\alpha(e_n)^2 - \mathbb{E}\,h_{n,\alpha}^2\,. \tag{3.6}$$

The last equality holds because $h_{n,\alpha} \in \mathcal{H}_{\mathrm{gr}(\alpha)}(X)$ exactly, so it is orthogonal to lower-degree terms in $\psi_\alpha(e_n)$, and the unique leading monomial of $\psi_\alpha(e_n)$ is the same as that of $h_{n,\alpha}$.

By Lemma 3.6, the norm of every proper flattening of every $e_{n,i}$ vanishes. By the fourth moment theorem (Theorem B.1), the $e_{n,i}$ supported in α are asymptotically jointly Gaussian (note that the proof of Theorem B.1 still holds when all convergence is along $\mathcal{U}$). Then we have the convergence of the variance $\mathbb{E}\,\psi_\alpha(e_n)^2 \underset{n\to\mathcal{U}}{\longrightarrow} 1$ by Lemma A.4. We have $\mathbb{E}\,h_{n,\alpha}^2 \underset{n\to\mathcal{U}}{\longrightarrow} 1$ by the combinatorial moment calculation in Lemma B.6. Therefore, Eq. (3.6) goes to 0 along $\mathcal{U}$ which proves the claim. □

Proof of Theorem 3.5. We observed above that a spanning set for $\mathscr{H}_d$ is

$$\left\{\frac{1}{\sqrt{\alpha!}} \bigdiamond_{i\in I} [e_{n,i}^{\diamond\alpha_i}] : \alpha \in \mathbb{N}^I, \mathrm{gr}(\alpha) = d\right\}.$$

By Lemma 3.7, these equal $\{[\psi_\alpha(e_n)] : \alpha \in \mathbb{N}^I, \mathrm{gr}(\alpha) = d\}$. The Gaussian convergence in Theorem B.1 implies the $[\psi_\alpha(e_n)]$ are orthonormal in $\mathscr{H}_d$ like the usual Hermite basis.

For the second part of the theorem, if the basis representation of $[p_{n,k}]$ is finite, then convergence in distribution follows from Theorem B.1 and the continuous mapping theorem. It remains to handle the case where the basis representation of $[p_{n,k}]$ has countably many nonzero terms. The expression

for p_k has $\mathbb{E}\, p_k^2 < \infty$ (hence it is well-defined in $L_2(Z)$) because, by orthonormality in $\mathcal{H}_{d_k}^{\mathrm{gr}}(Z)$ and $\mathscr{H}_{d_k}$:

$$\mathbb{E}\, p_k^2 = \sum_{\alpha \in \mathbb{N}^I : \mathrm{gr}(\alpha) = d_k} c_{k,\alpha}^2 = \lim_{n \to \mathcal{U}} \mathbb{E}\, p_{n,k}^2 < \infty\,. \tag{3.7}$$

Fix $\varepsilon > 0$ and a finite set F_k such that $\sum_{\alpha \notin F_k} c_{k,\alpha}^2 \le \varepsilon$ and define the truncated approximations,

$$[q_{n,k}] = \sum_{\alpha \in F_k} c_{k,\alpha} [\psi_\alpha(e_n)], \qquad q_k = \sum_{\alpha \in F_k} c_{k,\alpha} \psi_\alpha(Z)\,.$$

We have simultaneously:

$$\mathbb{E}(p_k - q_k)^2 = \sum_{\alpha \notin F_k} c_{k,\alpha}^2 \le \varepsilon \qquad\qquad \lim_{n \to \mathcal{U}} \mathbb{E}(p_{n,k} - q_{n,k})^2 = \sum_{\alpha \notin F_k} c_{k,\alpha}^2 \le \varepsilon \tag{3.8}$$

Furthermore, since F_k is finite, we have $(q_{n,1}, \dots, q_{n,k}) \xrightarrow[n \to \mathcal{U}]{d} (q_1, \dots, q_k)$ by the finite case.

To prove $(p_{n,k})_{k \in \mathbb{N}} \xrightarrow[n \to \mathcal{U}]{d} (p_k)_{k \in \mathbb{N}}$ it is equivalent to prove $(p_{n,1}, \dots, p_{n,k}) \xrightarrow[n \to \mathcal{U}]{d} (p_1, \dots, p_k)$ for each k. Fix a bounded 1-Lipschitz test function $\varphi : \mathbb{R}^k \to \mathbb{R}$. Let $P_n = (p_{n,1}, \dots, p_{n,k}), P = (p_1, \dots, p_k), Q_n = (q_{n,1}, \dots, q_{n,k}), Q = (q_1, \dots, q_k)$. Non-asymptotically,

$$\begin{aligned}
&|\,\mathbb{E}\,\varphi(P_n) - \mathbb{E}\,\varphi(P)| \\
&\le |\,\mathbb{E}\,\varphi(P_n) - \mathbb{E}\,\varphi(Q_n)| + |\,\mathbb{E}\,\varphi(Q_n) - \mathbb{E}\,\varphi(Q)| + |\,\mathbb{E}\,\varphi(Q) - \mathbb{E}\,\varphi(P)| \\
&\le \left(\sum_{i=1}^k \mathbb{E}(p_{n,i} - q_{n,i})^2\right)^{1/2} + |\,\mathbb{E}\,\varphi(Q_n) - \mathbb{E}\,\varphi(Q)| + \left(\sum_{i=1}^k \mathbb{E}(p_i - q_i)^2\right)^{1/2} \qquad \text{(1-Lipschitz)}
\end{aligned}$$

The right term is at most $\sqrt{k\varepsilon}$ by Eq. (3.8). In the limit $n \to \mathcal{U}$, the left term is at most $\sqrt{k\varepsilon}$ and the middle term converges to 0 since $Q_n \xrightarrow[n \to \mathcal{U}]{d} Q$. We conclude, for every $\varepsilon > 0$,

$$\lim_{n \to \mathcal{U}} |\,\mathbb{E}\,\varphi(P_n) - \mathbb{E}\,\varphi(P)| \le 2\sqrt{k\varepsilon}\,.$$

Take $\varepsilon \downarrow 0$ to get $\lim_{n \to \mathcal{U}} |\,\mathbb{E}\,\varphi(P_n) - \mathbb{E}\,\varphi(P)| = 0$. Bounded 1-Lipschitz functions determine weak convergence by the Portmanteau theorem, so $P_n \xrightarrow[n \to \mathcal{U}]{d} P$. □

3.4 Proof of regularity lemmas

Let $p \in \mathcal{H}_d(X), \mathbb{E}\, p^2 \le 1$ with Hermite coefficient tensor P. The idea of the regularity lemma is, if a flattening $P^{(r)}$ has a large singular value σ, then write

$$\begin{aligned}
P^{(r)} &= \sigma U^{(r)} (V^{(d-r)})^\intercal + P_{\mathrm{rem}}^{(r)}, && (U^{(r)} \in \mathbb{R}^{n^r}, V^{(d-r)} \in \mathbb{R}^{n^{d-r}}) \\
p &= \frac{\sigma}{c_{r,d-r}} u \diamond v + p_{\mathrm{rem}}, && (u \in \mathcal{H}_{\le r}(X), v \in \mathcal{H}_{\le d-r}(X))\,.
\end{aligned}$$

$U^{(r)}$ and $V^{(d-r)}$ are the (flattenings of the) Hermite coefficient tensors for lower-degree polynomials u and v. Since u and v are lower degree, they can be decomposed by induction.

On the other hand, removing the singular direction decreases the squared norm,[2]

$$\mathbb{E}\, p_{\mathrm{rem}}^2 \le \|P_{\mathrm{rem}}^{(r)}\|_F^2 = \|P^{(r)}\|_F^2 - \sigma^2 = \|P\|_F^2 - \sigma^2 = \mathbb{E}\, p^2 - \sigma^2\,.$$

Since initially $\mathbb{E}\, p^2 \le 1$, after removing at most $1/\varepsilon^2$ singular directions, all proper flattenings have spectral norm at most ε. Once all proper flattenings are small, by Theorem B.1, the remaining polynomial is Gaussian up to error depending on ε.

The ultrapower is essentially carrying out this approach implicitly, without having to worry about the recursion or the quantitative parameters. The energy measure $\mathbb{E}\, p^2 = \|P\|_F^2$ and the recursive decomposition are replaced by orthogonal decomposition in the Hilbert space $\mathscr{H}_d$. We now use this approach to prove the regularity lemmas.

Theorem 3.8 (Regularity lemma, compactification version)**.** *Let Z be a set of graded variables with countably many variables of each grade. Let $d_k \in \mathbb{N}$ for $k \in \mathbb{N}$. Let $p_{n,k} \in \mathcal{H}^{\mathrm{gr}}_{d_k}(Z)$ such that $\sup_n \mathbb{E}\, p_{n,k}^2 < \infty$ for each $k \in \mathbb{N}$. Then there exist $p_k \in \mathcal{H}^{\mathrm{gr}}_{d_k}(Z)$ and a subsequence of n such that $(p_{n,k})_{k\in\mathbb{N}} \xrightarrow{d} (p_k)_{k\in\mathbb{N}}$ along the subsequence. Consequently, $\{\mathrm{Law}(p_1,\dots,p_k) : p_i \in \mathcal{H}^{\mathrm{gr}}_{d_i}(Z), \mathbb{E}\, p_i^2 \le 1\}$ is weakly compact for each k.*

Proof. First we show that, without loss of generality, we may assume $p_{n,k}$ are ordinary Gaussian polynomials (all variables have grade one). By Lemma 2.1 applied to $(p_{n,1},\dots,p_{n,n})$, also using that bounded 1-Lipschitz test functions determine convergence in distribution by the Portmanteau theorem, there exist ordinary polynomials $q_{n,k}$ such that, for all $k \le n$,

$$|\mathbb{E}\, \varphi(p_{n,1},\dots,p_{n,k}) - \mathbb{E}\, \varphi(q_{n,1},\dots,q_{n,k})| \le \frac{1}{n}$$

for all bounded 1-Lipschitz functions $\varphi : \mathbb{R}^k \to \mathbb{R}$. So, if a subsequence of $q_{n,k}$ converges in distribution, then the same subsequence of $p_{n,k}$ converges in distribution to the same limit. By Lemma A.4, the approximants $q_{n,i}$ also satisfy $\sup_n \mathbb{E}\, q_{n,i}^2 < \infty$. Hence, it suffices to assume $p_{n,k} \in \mathcal{H}_{d_k}(X_1,\dots,X_n)$ are Gaussian polynomials such that $\sup_n \mathbb{E}\, p_{n,k}^2 < \infty$.

Applying Theorem 3.5, there exist graded Gaussian polynomials $p_k \in \mathcal{H}^{\mathrm{gr}}_{d_k}(Z)$ such that

$$(p_{n,k})_{k\in\mathbb{N}} \xrightarrow[n\to\mathcal{U}]{d} (p_k)_{k\in\mathbb{N}} \tag{3.9}$$

Each p_k is a countable or finite sum of Hermite polynomials in the Z variables, so the entire family $(p_k)_{k\in\mathbb{N}}$ can be represented using at most countably many graded Gaussian variables.

The weak topology on probability measures on $\mathbb{R}^{\mathbb{N}}$ is metrizable (by the sup over bounded 1-Lipschitz functions weighted by 2^{-i} in coordinate i), so we may use Lemma 3.2(iii) to extract a subsequence such that Eq. (3.9) holds ordinarily.

For the final assertion, apply this result to any sequence of admissible $(p_{n,1},\dots,p_{n,k})$. The Portmanteau theorem gives, along the extracted subsequence,

$$\mathbb{E}\, p_i^2 \le \liminf_{n\to\infty} \mathbb{E}\, p_{n,i}^2 \le 1.$$

Hence p_i is also admissible, and the set of admissible laws is compact. □

[2]The first inequality is not an equality because the Hermite coefficient tensor of p_{rem} is not P_{rem} but the symmetrization of P_{rem}. Symmetrization cannot increase the Frobenius norm.

Theorem 3.9 (Regularity lemma, approximation version)**.** *For every $k, D \in \mathbb{N}$, $\varepsilon > 0$ there exists $M = M(k, D, \varepsilon) \in \mathbb{N}$ such that the following holds. Let $d_1, \ldots, d_k \leq D$ (repeats allowed). Let $p_i \in \mathcal{H}_{d_i}(X)$ such that $\mathbb{E}\, p_i^2 \leq 1$ for all $i = 1, \ldots, k$. Then there exist $m \leq M, g_a \leq D$ and $Z_a \in \mathcal{H}_{g_a}(X)$ for $a \in [m]$, and $q_i \in \mathcal{H}_{d_i}^{\mathrm{gr}}(Z_1, \ldots, Z_m)$ where Z_a has grade g_a, such that:*

(i) $\sum_{i=1}^k \mathbb{E}(p_i - q_i)^2 \leq \varepsilon$

(ii) $W_2((Z_1, \ldots, Z_m), \mathcal{N}(0,1)^{\otimes m}) \leq \varepsilon$

Proof. Suppose the assertion fails for some k, D, ε. Then for every M, there is a choice of polynomials which require at least M variables to approximate within error 4ε. Construct a sequence $p_{n,i} \in \mathcal{H}_{d_i}(X), \mathbb{E}\, p_{n,i}^2 \leq 1$ such that the number of Gaussians required to approximate them within error 4ε grows with n. We will use the ultraproduct to approximate an infinite subsequence using a constant number of variables, yielding the contradiction.

Strictly speaking, the $p_{n,i}$ may not be polynomials since X could be infinite. For each one, truncate the Hermite expansion to a finite approximant within ε error. Then we obtain a sequence of bona fide polynomials $p_{n,i} \in \mathcal{H}_{d_i}(X_1, \ldots, X_n), \mathbb{E}\, p_{n,i}^2 \leq 1$ such that the number of Gaussians required to approximate them within error ε grows with n. Embed $[p_{n,i}]$ into $\mathscr{H}_{d_i}$.

By Theorem 3.5, write the basis representation

$$[p_{n,i}] = \sum_{\mathrm{gr}(\alpha) = d_i} c_{i,\alpha} [\psi_\alpha(e_n)].$$

Choose finite sets F_i of multi-indices of grade d_i such that $\sum_{i=1}^k \sum_{\alpha \notin F_i} c_{i,\alpha}^2 < \varepsilon/2$. List the finitely many primitive elements $[e_{n,1}], \ldots, [e_{n,m}]$ appearing in the union of the F_i and for each one select a representative sequence of polynomials, written $Z_{n,a} \in \mathcal{H}_{g_a}(X_1, \ldots, X_n)$.

Define the graded-degree-d_i polynomial $q_i(Z_1, \ldots, Z_m) = \sum_{\alpha \in F_i} c_{i,\alpha} \psi_\alpha(Z)$. By asymptotic orthonormality of the $\psi_\alpha(Z)$ in the ultraproduct,

$$\lim_{n \to \mathcal{U}} \sum_{i=1}^k \mathbb{E}\big(p_{n,i} - q_i(Z_{n,1}, \ldots, Z_{n,m})\big)^2 = \sum_{i=1}^k \sum_{\substack{\mathrm{gr}(\alpha) = d_i \\ \alpha \notin F_i}} c_{i,\alpha}^2 < \varepsilon/2. \tag{3.10}$$

By Lemma 3.2(i), inequality (3.10) holds with error ε for all n in a $\mathcal{U}$-large set. Call this set A.

The second part of Theorem 3.5 gives $(Z_{n,1}, \ldots, Z_{n,m}) \xrightarrow[n \to \mathcal{U}]{d} \mathcal{N}(0,1)^{\otimes m}$. We claim that the following set B is also $\mathcal{U}$-large,

$$B = \{n : W_2((Z_{n,1}, \ldots, Z_{n,m}), \mathcal{N}(0,1)^{\otimes m}) < \varepsilon\}.$$

Otherwise, $\mathbb{N} \setminus B \in \mathcal{U}$. By Lemma 3.2(iii) there is a subsequence of $n \in \mathbb{N} \setminus B$ such that $(Z_{n,1}, \ldots, Z_{n,m}) \xrightarrow{d} \mathcal{N}(0,1)^{\otimes m}$ along the subsequence. By Lemma A.4, this also implies convergence in W_2, contradicting the definition of B. Thus $B \in \mathcal{U}$.

Then $A \cap B$ is also $\mathcal{U}$-large. Use Lemma 3.2(iii) to obtain a subsequence of $p_{n,i}$ which are approximated up to error ε by at most m variables, the desired contradiction. □

Remark. The proof of Theorem 3.9 gives a nonconstructive bound on $M(k, D, \varepsilon)$. GPT-6 Astra asserts that the strategy described at the start of this section can prove an Ackermann-type recursion for $M(k, D, \varepsilon)$, for example, $M(k, 2, \varepsilon) \leq Ck^4 \varepsilon^{-3} (k+1)^{Ck/\varepsilon}$, a tower-type bound for $D = 3$, and larger bounds for $D > 3$. We have not attempted to verify the proof.

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A Probability preliminaries

Lemma A.1 ([Bil99, Example 2.4])**.** *Let $x_{n,k}$ be real-valued random variables. Endow $\mathbb{R}^{\mathbb{N}}$ with the product topology. The following are equivalent:*

(i) $(x_{n,k})_{k\in\mathbb{N}} \xrightarrow{d} (x_k)_{k\in\mathbb{N}}$

(ii) $(x_{n,1},\dots,x_{n,k}) \xrightarrow{d} (x_1,\dots,x_k)$ *for all* k

(iii) $(x_{n,j})_{j\in J} \xrightarrow{d} (x_j)_{j\in J}$ *for all finite sets* $J \subseteq \mathbb{N}$.

Lemma A.2 (Method of moments [Bil95, Theorems 29.4, 30.1, 30.2])**.** *Let $x_{n,k}$ be real-valued random variables. Assume that $(x_1,\dots,x_k)$ have a jointly Gaussian distribution. Assume for all finitely supported $\alpha \in \mathbb{N}^k$,*

$$\mathbb{E}\prod_{i=1}^{k} x_{n,i}^{\alpha_i} \longrightarrow \mathbb{E}\prod_{i=1}^{k} x_i^{\alpha_i}\,.$$

Then $(x_{n,1},\dots,x_{n,k}) \xrightarrow{d} (x_1,\dots,x_k)$.

Lemma A.3 (Gaussian hypercontractive inequality)**.** *For all $p \in \mathcal{H}_{\le d}(X)$ and $q \ge 2$,*

$$\|p\|_q \le (q-1)^{d/2}\|p\|_2\,.$$

Lemma A.4. *Let $d_1, \dots, d_k \in \mathbb{N}$ and $p_{n,i} \in \mathcal{H}_{\le d_i}(X)$. If $(p_{n,1}, \dots, p_{n,k}) \xrightarrow{d} (p_1, \dots, p_k)$ then all mixed moments converge, and the convergence also holds in W_p for $p \in [1, \infty)$. The same holds for graded polynomials $p_{n,i} \in \mathcal{H}^{\mathrm{gr}}_{d_i}(Z)$.*

Proof. Write $P_n = (p_{n,1}, \dots, p_{n,k})$ and $P = (p_1, \dots, p_k)$. Since $P_n \xrightarrow{d} P$, each coordinate sequence $(p_{n,i})_{n \in \mathbb{N}}$ has tight laws. Recall a collection of probability measures on $\mathbb{R}^I$ is *tight* if for every $\varepsilon > 0$ there is compact $K \subseteq \mathbb{R}^I$ such that $\mu(K) \ge 1 - \varepsilon$ for every measure μ in the collection.

By tightness, the Paley–Zygmund inequality, and Lemma A.3, we have for $p_{n,k} \neq 0$,

$$\Pr\left(|p_{n,k}| \ge \frac{\|p_{n,k}\|_2}{\sqrt{2}}\right) \ge \frac{\|p_{n,k}\|_2^4}{4\|p_{n,k}\|_4^4} \ge \frac{1}{4 \cdot 9^{d_k}}.$$

Thus tightness implies $\sup_n \|p_{n,k}\|_2 < \infty$. Applying hypercontractivity again, for every $q \ge 2$,

$$\sup_n \|p_{n,k}\|_q \le (q-1)^{d_k/2} \sup_n \|p_{n,k}\|_2 < \infty.$$

Consequently, we deduce $\sup_n \mathbb{E}\,\|P_n\|^q < \infty$ and, by the Portmanteau theorem, $\mathbb{E}\,\|P\|^q < \infty$ for all $q \ge 1$. Then for every continuous function φ of polynomial growth, the sequence $(\varphi(P_n))_{n \in \mathbb{N}}$ is uniformly integrable. The continuous mapping theorem and [Bil99, Theorem 3.5] imply

$$\mathbb{E}\,\varphi(P_n) \longrightarrow \mathbb{E}\,\varphi(P).$$

Thus we have convergence of all mixed moments. Taking $\varphi(x) = \|x\|^q$ gives convergence of the qth absolute moments. Together with weak convergence, this implies $W_q(\mathrm{Law}(P_n), \mathrm{Law}(P)) \to 0$ by [Vil09, Theorem 6.9]. Forgetting grades shows $\mathcal{H}^{\mathrm{gr}}_d(Z) \subseteq \mathcal{H}_{\le d}(Z)$ so the same argument applies to graded polynomials. □

B Combinatorial proof of the fourth moment theorem

We use the tensor notation from Section 3.3.1. We adopt the Nualart and Peccati [NP05] notation that $P \otimes_r P$ equals $(P^{(r)})^\intercal P^{(r)}$ unflattened into a tensor. $P \otimes_r P$ is a non-symmetric tensor with order $2(d - r)$ when P has order d.

Theorem B.1 (Fourth moment theorem [NP05, NOL08])**.** *Let $d_1, \dots, d_k \in \mathbb{N}$ (repeats allowed). Let $p_{n,i} \in \mathcal{H}_{d_i}(X_1, \dots, X_n)$. Let $P_{n,i}$ be their Hermite coefficient tensors. Assume that $\lim_{n\to\infty} \mathbb{E}\, p_{n,i} p_{n,j} = \Sigma_{ij}$ for some matrix $\Sigma \in \mathbb{R}^{k \times k}$. The following are equivalent:*

(i) $(p_{n,1}, \dots, p_{n,k}) \xrightarrow{d} \mathcal{N}(0, \Sigma)$

(ii) $\mathbb{E}\, p_{n,i}^4 \to 3\Sigma_{ii}^2$ *for all* $i = 1, \dots, k$.

(iii) $\|P_{n,i} \otimes_r P_{n,i}\|_F \to 0$ *for all* $0 < r < d_i$ *and* $i = 1, \dots, k$.

(iv) $\|P_{n,i}^{(r)}\| \to 0$ *for all* $0 < r < d_i$ *and* $i = 1, \dots, k$.

Equivalence of (i),(ii),(iii) was proven by [NP05, NOL08]. It is easy to check that (iii) is equivalent to (iv) using the relations:

$$\|P^{(r)}\|^2 \le \|P \otimes_r P\|_F \le \|P^{(r)}\| \|P\|_F\,. \tag{B.1}$$

To see these relations, let s_j be the singular values of $P^{(r)}$. Eq. (B.1) follows from the equations,

$$\|P^{(r)}\| = \max_j s_j, \qquad \|P \otimes_r P\|_F^2 = \sum_j s_j^4, \qquad \|P\|_F^2 = \sum_j s_j^2 .$$

Since $\|P_{n,i}\|_F^2 = \mathbb{E}\, p_{n,i}^2$ converges to the constant Σ_{ii}, then Eq. (B.1) implies $\|P \otimes_r P\|_F$ vanishes if and only if $\|P_{n,i}^{(r)}\|$ vanishes. Hence (iii) is equivalent to (iv).

We now give a direct proof of Theorem B.1 using the moment method (see also [Nou11] for a prior combinatorial proof). The moments of Hermite polynomials have natural combinatorial interpretations in terms of matchings.

Lemma B.2 ([Jan97, Chapter 3])**.** *Let $\alpha^1, \dots, \alpha^\ell \in \mathbb{N}^I$ be finitely supported. Then:*

$$\mathbb{E}\left[\prod_{j=1}^{\ell} \mathrm{He}_{\alpha^j}(X)\right] = \sum_{M \in \mathcal{M}(\alpha^1,\dots,\alpha^\ell)} \prod_{uv \in M} \mathbf{1}_{u=v}$$

where we take ℓ groups of objects, with the jth group being a multiset of size $|\alpha^j|$ with α_i^j copies of each $i \in I$. Let $\mathcal{M}(\alpha^1, \dots, \alpha^\ell)$ be the set of perfect matchings on the disjoint union of the groups such that no two objects are matched within a group. The equality $u = v$ is on I.

A *tensor network* is a multigraph π along with a collection of symmetric tensors $T_a \in (\mathbb{R}^n)^{\otimes \deg(a)}_{\mathrm{sym}}$ associated to the vertices of π. The value of the tensor network is:

$$\mathrm{val}_\pi(T) = \sum_{\varphi : E(\pi) \to [n]} \prod_{a \in V(\pi)} (T_a)_{\varphi(a)}$$

where $\varphi(a)$ is the multiset of labels on the edges incident to a (the order of the incident edges does not matter since T_a is assumed to be symmetric).

Lemma B.3. *Let (T, π) be a tensor network such that π has no self-loops. Then:*

$$|\,\mathrm{val}_\pi(T)| \le \prod_{a \in V(\pi)} \|T_a\|_F .$$

Proof. $\mathrm{val}_\pi(T)$ factors across connected components of π, so we may assume π is connected. Use induction on $|V(\pi)|$. We also upgrade the statement to allow non-symmetric tensors (in this case, each vertex of π comes with an ordering of the incident edges).

In the base case, π is a single multiedge between two tensors R, S,

$$|\mathrm{val}_\pi(T)| = \left|\sum_{i \in [n]^d} R_i S_i\right| \le \left(\sum_{i \in [n]^d} R_i^2\right)^{1/2} \left(\sum_{i \in [n]^d} S_i^2\right)^{1/2} = \|R\|_F \|S\|_F .$$

In the induction step, let a, b be the endpoints of a multiedge in π with associated tensors R, S. Let π' be π with the multiedge contracted. Let Q be the contraction of R, S along the multiedges between a, b and associate Q to the contracted vertex in π'. Call this new tensor network (T', π'). Then $\mathrm{val}_\pi(T) = \mathrm{val}_{\pi'}(T')$ by definition. By induction,

$$|\mathrm{val}_{\pi'}(T')| \le \|Q\|_F \prod_{c \in V(\pi) \setminus \{a,b\}} \|T_c\|_F .$$

We can express Q as a flattened matrix multiplication $R^{(L_a,R_a)}S^{(L_b,R_b)}$ followed by unflattening, where R_a, L_b are the edges between a, b. Using $\|AB\|_F \le \|A\|_F\|B\|_F$ on the matrix multiplication,

$$\|Q\|_F \prod_{c\in V(\pi)\setminus\{a,b\}} \|T_c\|_F \le \|R\|_F\|S\|_F \prod_{c\in V(\pi)\setminus\{a,b\}} \|T_c\|_F = \prod_{a\in V(\pi)} \|T_a\|_F\,. \qquad \square$$

Lemma B.4. *Let $d_1,\dots,d_k \ge 1$. Let $p_a \in \mathcal{H}_{d_a}(X_1,\dots,X_n)$ such that $\mathbb{E}\,p_a^2 \le 1$ and let P_a be their Hermite coefficient tensors. Assume that $\|P_a^{(r)}\| \le \delta$ for all $a \in [k]$ and $0 < r < d_a$. Let $Z = (Z_1,\dots,Z_k)$ be a centered Gaussian vector with the same covariance matrix as $(p_1,\dots,p_k)$, and let $D = \sum_{a=1}^k d_a$. If D is even, then*

$$\left|\mathbb{E}\prod_{a=1}^k p_a - \mathbb{E}\prod_{a=1}^k Z_a\right| \le \frac{(D-1)!!}{\sqrt{d_1!\cdots d_k!}}\,\delta. \tag{B.2}$$

If D is odd, both expectations vanish.

Proof. Take a set of k vertices, where a is incident to d_a labeled half-edge endpoints. Let $\mathcal{D}$ be the set of perfect matchings of all D half-edges such that no two half-edges at the same vertex are matched. $\pi \in \mathcal{D}$ forms a tensor network with $P_1,\dots,P_k$. We have,

$$\begin{aligned}\mathbb{E}\prod_{a=1}^k p_a &= \frac{1}{\sqrt{d_1!\cdots d_k!}} \sum_{i_a\in[n]^{d_a}} \prod_{a=1}^k (P_a)_{i_a}\,\mathbb{E}\left[\prod_{a=1}^k \mathrm{He}_{i_a}(X)\right] && \text{(Eq. (3.3))}\\ &= \frac{1}{\sqrt{d_1!\cdots d_k!}} \sum_{\pi\in\mathcal{D}} \mathrm{val}_\pi(P_1,\dots,P_k). && \text{(Lemma B.2)} \end{aligned} \tag{B.3}$$

Eq. (B.3) is a "Feynman diagram expansion" for the moment $\mathbb{E}\prod_{a=1}^k p_a$. When D is odd, $\mathcal{D}$ is empty, and the expectation is zero as needed.

Call a diagram π a *pair diagram* if every connected component has exactly two vertices. Such a component on vertices a, b is possible only if $d_a = d_b = d$. Its $d!$ possible matchings all give $\langle P_a, P_b\rangle_F = \mathbb{E}[p_a p_b]$, and its normalization in (B.3) is $1/d!$. Thus the sum over pair diagrams equals

$$\sum_{\text{pairings } \eta \text{ of } \{1,\dots,k\}} \prod_{\{a,b\}\in\eta} \mathbb{E}[p_a p_b] = \mathbb{E}\prod_{a=1}^k Z_a.$$

Every remaining diagram has a connected component with at least three vertices. Choose adjacent vertices a, b in this component, and let $r \ge 1$ be the number of edges between them. These edges cannot be all edges incident to both vertices, so, after interchanging a, b if necessary, $0 < r < d_a$. As in the proof of Lemma B.3, form a new tensor network π' by contracting the vertices a, b and putting at the contracted vertex the tensor Q which is the contraction of P_a, P_b. This defines $\mathrm{val}_\pi(P) = \mathrm{val}_{\pi'}(P, Q)$. By Lemma B.3,

$$|\mathrm{val}_{\pi'}(P,Q)| \le \|Q\|_F \prod_{c\in[k]\setminus\{a,b\}} \|P_c\|_F\,.$$

We can express Q as the unflattening of a matrix multiplication $(P_a^{(r)})^\intercal P_b^{(r)}$. Use $\|AB\|_F \le \|A\|\|B\|_F$ on the matrix multiplication to deduce,

$$\|Q\|_F \prod_{c\in[k]\setminus\{a,b\}} \|P_c\|_F \le \|P_a^{(r)}\| \prod_{b\in[k]\setminus\{a\}} \|P_b\|_F \le \|P_a^{(r)}\| \le \delta\,.$$

The second-to-last inequality uses $\|P_b\|_F^2 = \mathbb{E}\, p_b^2 \le 1$. Thus $|\mathrm{val}_\pi(P)| \le \delta$ for each non-pair diagram in Eq. (B.3). There are at most $(D-1)!!$ matchings of D slots, which proves Eq. (B.2). □

Proof of Theorem B.1. (i) $\implies$ (ii). Convergence in distribution implies convergence of moments by Lemma A.4.

(ii) $\implies$ (iii). Let $p \in \mathcal{H}_d(X_1, \dots, X_n)$ have Hermite coefficient tensor P. The fourth cumulant $\kappa_4 := \mathbb{E}\, p^4 - 3(\mathbb{E}\, p^2)^2$ can be written as a sum-of-squares expression in terms of tensor contractions as [NP05, Proof of Proposition 3, (i) $\implies$ (ii)],

$$\kappa_4 = \sum_{r=1}^{d-1} \binom{d}{r}^2 \left(\|P \otimes_r P\|_F^2 + \binom{2d-2r}{d-r} \|\operatorname{Sym}(P \otimes_r P)\|_F^2 \right). \tag{B.4}$$

Therefore, if $\kappa_4 \to 0$ as $n \to \infty$ then all of the tensor norms $\|P \otimes_r P\|_F$ must also vanish.

Here is a combinatorial interpretation of (B.4). Following Eq. (B.3),

$$\mathbb{E}\, p^4 = \frac{1}{(d!)^2} \sum_{\pi \in \mathcal{D}} \mathrm{val}_\pi(P, P, P, P)$$

where $\mathcal{D}$ is the set of perfect matchings on four vertices with d incident half-edges such that no half-edges are matched at the same vertex.

Suppose π has r edges matched between vertex 1 and 2. Then there are also r edges matched between vertex 3 and 4 and there are $\left(\binom{d}{r}^2 r!\right)^2$ ways to choose this part of π. We interpret the edges between 1,2 as contracted in $P \otimes_r P$ and the edges outgoing from 1,2 as the indices of $P \otimes_r P$.

There are $(2d-2r)!$ ways to choose the remaining part of π. If the remaining part of π matches the half-edges at 1 with the half-edges at 3 (i.e. π is a cycle graph), we obtain exactly $\mathrm{val}_\pi(P, P, P, P) = \|P \otimes_r P\|_F^2$. However, in general, the sum over π treats all outgoing edges from $\{1, 2\}$ in the same way, so the average value of $\mathrm{val}_\pi(P, P, P, P)$ is $\|\operatorname{Sym}(P \otimes_r P)\|_F^2$. In total,

$$\begin{aligned} \frac{1}{(d!)^2} \sum_{\pi \in \mathcal{D}} \mathrm{val}_\pi(P, P, P, P) &= \sum_{r=0}^{d} \frac{(2d-2r)!}{(d!)^2} \left(\binom{d}{r}^2 r! \right)^2 \|\operatorname{Sym}(P \otimes_r P)\|_F^2 \\ &= \sum_{r=0}^{d} \binom{d}{r}^2 \binom{2d-2r}{d-r} \|\operatorname{Sym}(P \otimes_r P)\|_F^2 \end{aligned} \tag{B.5}$$

We re-analyze the terms $r = 0$ and $r = d$ separately. The term $r = d$ equals $\|P\|_F^4 = (\mathbb{E}\, p^2)^2$. All π in the term $r = 0$ are cyclic graphs (or disconnected). Let s be the number of 1-3 edges which equals the number of 2-4 edges. There are $\binom{d}{s}^4 (s!)^2((d-s)!)^2 = \binom{d}{s}^2 (d!)^2$ choices for π. For these cases, $\mathrm{val}_\pi(P, P, P, P) = \|P \otimes_s P\|_F^2$. Therefore,

$$\begin{aligned} \frac{1}{(d!)^2} \sum_{\pi \in \mathcal{D}} \mathrm{val}_\pi(P, P, P, P) &= \|P\|_F^4 + \sum_{s=0}^{d} \binom{d}{s}^2 \|P \otimes_s P\|_F^2 + \sum_{r=1}^{d-1} \binom{d}{r}^2 \binom{2d-2r}{d-r} \|\operatorname{Sym}(P \otimes_r P)\|_F^2 \\ &= 3\|P\|_F^4 + \sum_{s=1}^{d-1} \binom{d}{s}^2 \|P \otimes_s P\|_F^2 + \sum_{r=1}^{d-1} \binom{d}{r}^2 \binom{2d-2r}{d-r} \|\operatorname{Sym}(P \otimes_r P)\|_F^2 \end{aligned}$$

Recall this equals $\mathbb{E}\, p^4$. Subtract $3\|P\|_F^4 = 3(\mathbb{E}\, p^2)^2$ to obtain κ_4.

(iii) $\Longrightarrow$ (iv). Proven immediately after the theorem statement.

(iv) $\Longrightarrow$ (i). Let Z_n be a centered Gaussian vector with the same covariance matrix as $(p_{n,1}, \dots, p_{n,k})$. The variance of Z_n is uniformly bounded since $\mathbb{E}\, p_{n,i}^2 \to \Sigma_{ii}$, say by a constant $C \geq 1$. We may apply Lemma B.4 to get, for every $(m_1, \dots, m_k) \in \mathbb{N}^k$, letting $|m| = \sum_a m_a$ and $D = \sum_a m_a d_a$ and $\delta_n = \max_{a\in[k]} \max_{0<r<d_a} \|P_{n,a}^{(r)}\|$,

$$|\mathbb{E} \prod_{a=1}^k p_{n,a}^{m_a} - \mathbb{E} \prod_{a=1}^k Z_{n,a}^{m_a}| \leq C^{|m|}(D-1)!!\, \delta_n \longrightarrow 0.$$

The covariances and hence also all moments of Z_n converge to those of $\mathcal{N}(0, \Sigma)$. So by the method of moments in Lemma A.2, $(p_{n,1}, \dots, p_{n,k}) \xrightarrow{d} \mathcal{N}(0, \Sigma)$. □

B.1 Omitted moment calculations

Lemma B.5. *For $i, r \in \mathbb{N}$, let $S_{ir} = \frac{1}{\sqrt{N}} \sum_{j=1}^N \psi_r(G_{ir,j})$. For finitely-supported $\alpha \in \mathbb{N}^{\mathbb{N}\times\mathbb{N}}$, let*

$$U_{m,ir} = \frac{1}{N^{m/2}} \sum_{\substack{j_1,\dots,j_m\in[N] \\ j_1,\dots,j_m \text{ distinct}}} \psi_r(G_{ir,j_1})\psi_r(G_{ir,j_2}) \cdots \psi_r(G_{ir,j_m}), \qquad U_\alpha = \prod_{i,r} U_{\alpha_{ir},ir}\,.$$

Then for $G_{ir,j} \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$,

$$\mathbb{E}(U_\alpha - \mathrm{He}_\alpha(S))^2 \leq O_\alpha(N^{-1})\,.$$

Proof. Consider generic $S = N^{-1/2} \sum_{j=1}^N \psi_r(G_j)$ and $U_m = N^{-m/2} \sum_{j_1,\dots,j_m \text{ distinct}} \prod_{a=1}^m \psi_r(G_{j_a})$. Using Lemma B.2,

$$\mathbb{E}\, S^{2\ell} = \frac{1}{(Nr!)^\ell} \sum_{M\in\mathcal{M}(2\ell,r)} \sum_{\varphi:[2\ell]\to[N]} \prod_{uv\in M} \mathbf{1}_{\varphi(\text{group}(u))=\varphi(\text{group}(v))}\,.$$

where we have 2ℓ groups of r vertices each, $\mathcal{M}(2\ell, r)$ is the set of perfect matchings on the vertices such that no two vertices are matched within a group, and the indicator function checks whether all matched pairs use the same Gaussian G_k on their group. The inner summation is $N^{c(M)}$ where $c(M)$ is the number of connected components of M when the groups are contracted. Hence,

$$\mathbb{E}\, S^{2\ell} = \frac{1}{(Nr!)^\ell} \sum_{M\in\mathcal{M}(2\ell,r)} N^{c(M)} = (2\ell-1)!! + O_\alpha(N^{-1})\,. \tag{B.6}$$

The second equality separates the dominant terms where $\mathcal{M}$ is a perfect matching between groups from the remaining $O_\alpha(1)$ terms.

Express the Hermite polynomials in the monomial basis and use Eq. (B.6) to deduce $\mathbb{E}\,\mathrm{He}_m(S)^2 = \mathbb{E}\,\mathrm{He}_m(X)^2 + O_\alpha(N^{-1}) = m! + O_\alpha(N^{-1})$ for standard Gaussian variables X. We also have

$$\mathbb{E}\,\mathrm{He}_m(S)U_m = \mathbb{E}\,U_m^2 = m!\frac{(N)_m}{N^m} = m! + O_m(N^{-1})\,. \tag{B.7}$$

where $(N)_m$ is the falling factorial. Indeed, the first equality observes that the only component of $\mathrm{He}_m(S)$ which is not orthogonal to U_m is its "leading monomial" U_m itself. The second equality observes that nonzero terms in $\mathbb{E}\,U_m^2$ are perfect matchings between the indices in U_m. Therefore,

$$\mathbb{E}(\mathrm{He}_m(S) - U_m)^2 = \mathbb{E}\,\mathrm{He}_m(S)^2 - 2\,\mathbb{E}\,\mathrm{He}_m(S)U_m + \mathbb{E}\,U_m^2 = O_\alpha(N^{-1})\,.$$

Independence across the i, r coordinates implies the same holds for the multivariate case. □

Lemma B.6. *Let $d_1, \dots, d_k \geq 1$ and $D = \sum_{a=1}^k d_a$. Let $p_a \in \mathcal{H}_{d_a}(X_1, \dots, X_n)$ such that $\mathbb{E}\, p_a^2 = 1$ and let P_a be their Hermite coefficient tensors. Assume that $\|P_a^{(r)}\| \leq \delta$ for all $a \in [k]$ and $0 < r < d_a$. Then*

$$\left| \mathbb{E} \left(\bigdiamond_{a=1}^{k} p_a \right)^2 - \sum_{M \in \mathcal{M}(2,k)} \prod_{uv \in M} \mathbb{E}[p_u p_v] \right| \leq \frac{D!}{d_1! \cdots d_k!} \delta$$

where $\mathcal{M}(2,k)$ is the set of perfect matchings between two copies of $[k]$.

Proof sketch. The tensor $P = \sqrt{\frac{D!}{d_1! \cdots d_k!}} \otimes_{a=1}^k P_a$ represents the top product $\diamond_{a=1}^k p_a$. Then we write a diagrammatic expansion for $\mathbb{E} \left(\bigdiamond_{a=1}^{k} p_a \right)^2$ in the same way as Lemma B.4. Separating out the dominant and subdominant perfect matchings in the same way proves the claim. □